

Trigonometric Ratios

Practice Problem Set · Class 11 — JEE Foundation

Instructions: Evaluate each expression using the standard values of the angles 0° , 30° , 45° , 60° and 90° . Calculators are not permitted. (40 problems · 5 sections)

Section A — Direct substitution

1. $\sin 30^\circ + \cos 60^\circ$
2. $\tan 45^\circ + \cot 45^\circ$
3. $\sin 0^\circ + \tan 0^\circ + \cos 90^\circ$
4. $\sin 90^\circ \times \cos 0^\circ \times \tan 45^\circ$
5. $2 \sin 30^\circ \cdot \cos 60^\circ$
6. $\sin^2 30^\circ + \cos^2 60^\circ$
7. $\sin^2 60^\circ + \cos^2 30^\circ + \tan^2 45^\circ$
8. $\sin 60^\circ \cdot \cos 30^\circ - \cos 60^\circ \cdot \sin 30^\circ$

Section B — Multi-term standard-angle expressions

9. $2 \sin 30^\circ + \cos^2 60^\circ - \tan 45^\circ$
10. $\sin^2 45^\circ + \cos^2 45^\circ + \tan^2 60^\circ$
11. $\cos^2 30^\circ - \sin^2 30^\circ$
12. $4 \cot^2 45^\circ - \sec^2 60^\circ + \sin 90^\circ$
13. $2 \sin^2 30^\circ + 3 \cos^2 45^\circ + 3 \tan^2 30^\circ$
14. $(\sin 45^\circ + \cos 45^\circ)^2$
15. $(\cos 30^\circ + \sin 60^\circ)^2 + (\sin 30^\circ + \cos 60^\circ)^2$
16. $\sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$
17. $\cos^2 30^\circ + \cos^2 45^\circ + \cos^2 60^\circ$
18. $4(\sin^4 30^\circ + \cos^4 60^\circ) + 3(\cos^2 45^\circ + \sin^2 90^\circ)$
19. $2 \sin^2 30^\circ - 3 \cos^2 45^\circ + 4 \sin 90^\circ + 3 \cos 0^\circ$
20. $\operatorname{cosec}^2 30^\circ - 4 \cos^2 60^\circ \cdot \tan^2 45^\circ + 2 \sec^2 45^\circ$
21. $\cos^4 45^\circ + \sin^4 45^\circ$
22. $\sin^3 30^\circ + \cos^3 60^\circ$

Section C — Complementary-angle property

23. $\sin 25^\circ \cdot \cos 65^\circ + \cos 25^\circ \cdot \sin 65^\circ$
24. $\cos^2 20^\circ + \cos^2 70^\circ$
25. $\sin^2 54^\circ + \sin^2 36^\circ$
26. $\tan 35^\circ \cdot \tan 55^\circ$
27. $\sin 18^\circ / \cos 72^\circ$
28. $\sec 70^\circ \cdot \operatorname{cosec} 20^\circ - \tan 70^\circ \cdot \cot 20^\circ$
29. $\operatorname{cosec}^2 67^\circ - \tan^2 23^\circ$
30. $\sin 36^\circ / \cos 54^\circ + \sin 54^\circ / \cos 36^\circ$
31. $\sin^2 29^\circ + \sin^2 61^\circ$

Section D — Ratios from a right triangle

- 32.** In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 3$ cm, $BC = 4$ cm. Find $\sin A$, $\cos A$, $\tan A$ and $\sin C$.
- 33.** In $\triangle PQR$, $\angle Q = 90^\circ$, $PQ = 5$ cm, $PR = 13$ cm. Find $\tan P$ and $\tan R$.
- 34.** In $\triangle XYZ$, $\angle Y = 90^\circ$, $XY = 7$ cm, $XZ = 25$ cm. Find $\sin Z$ and $\cos Z$.
- 35.** In $\triangle ABC$, $\angle C = 90^\circ$, $AC = 15$ cm, $BC = 20$ cm. Find all six trigonometric ratios for angle A.

Section E — Challenge expressions

- 36.** $(\sin 30^\circ \cdot \cos 60^\circ + \cos 30^\circ \cdot \sin 60^\circ)^2 + (\cos 30^\circ \cdot \cos 60^\circ - \sin 30^\circ \cdot \sin 60^\circ)^2$
- 37.** $(1 + \cot^2 30^\circ) \times \sin^2 30^\circ$
- 38.** $(2 \tan 30^\circ) / (1 + \tan^2 30^\circ)$
- 39.** $(\tan 60^\circ - \tan 30^\circ) / (1 + \tan 60^\circ \cdot \tan 30^\circ)$
- 40.** $(1 - \tan^2 45^\circ) / (1 + \tan^2 45^\circ)$

Solutions

Final answers in **bold**. Identities used: $\sin(90^\circ - \theta) = \cos \theta$, $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$, $\sec^2 \theta - \tan^2 \theta = 1$.

Section A — Direct substitution

$$1. = \frac{1}{2} + \frac{1}{2} = \mathbf{1}$$

$$2. = 1 + 1 = \mathbf{2}$$

$$3. = 0 + 0 + 0 = \mathbf{0}$$

$$4. = 1 \times 1 \times 1 = \mathbf{1}$$

$$5. = 2 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

$$6. = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$7. = \frac{3}{4} + \frac{3}{4} + 1 = \mathbf{5/2}$$

$$8. = (\sqrt{3}/2)(\sqrt{3}/2) - (\frac{1}{2})(\frac{1}{2}) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

Section B — Multi-term standard-angle expressions

$$9. = 1 + \frac{1}{4} - 1 = \frac{1}{4}$$

$$10. = \frac{1}{2} + \frac{1}{2} + 3 = \mathbf{4}$$

$$11. = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$12. = 4(1) - 4 + 1 = \mathbf{1}$$

$$13. = 2(\frac{1}{4}) + 3(\frac{1}{2}) + 3(\frac{1}{3}) = \frac{1}{2} + \frac{3}{2} + 1 = \mathbf{3}$$

$$14. = (1/\sqrt{2} + 1/\sqrt{2})^2 = (\sqrt{2})^2 = \mathbf{2}$$

$$15. = (\sqrt{3})^2 + (1)^2 = 3 + 1 = \mathbf{4}$$

$$16. = \frac{1}{4} + \frac{1}{2} + \frac{3}{4} = \mathbf{3/2}$$

$$17. = \frac{3}{4} + \frac{1}{2} + \frac{1}{4} = \mathbf{3/2}$$

$$18. = 4(1/16 + 1/16) + 3(\frac{1}{2} + 1) = \frac{1}{2} + \frac{9}{2} = \mathbf{5}$$

$$19. = \frac{1}{2} - \frac{3}{2} + 4 + 3 = \mathbf{6}$$

$$20. = 4 - 4(\frac{1}{4}) + 2(2) = 4 - 1 + 4 = \mathbf{7}$$

$$21. = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$22. = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

Section C — Complementary-angle property

$$23. = \sin(25^\circ + 65^\circ) = \sin 90^\circ = \mathbf{1}$$

$$24. \cos 70^\circ = \sin 20^\circ \rightarrow \cos^2 20^\circ + \sin^2 20^\circ = \mathbf{1}$$

$$25. \sin 36^\circ = \cos 54^\circ \rightarrow \sin^2 54^\circ + \cos^2 54^\circ = \mathbf{1}$$

$$26. \tan 55^\circ = \cot 35^\circ \rightarrow \tan 35^\circ \cdot \cot 35^\circ = \mathbf{1}$$

$$27. \cos 72^\circ = \sin 18^\circ \rightarrow \sin 18^\circ / \sin 18^\circ = \mathbf{1}$$

$$28. \operatorname{cosec} 20^\circ = \sec 70^\circ, \cot 20^\circ = \tan 70^\circ \rightarrow \sec^2 70^\circ - \tan^2 70^\circ = \mathbf{1}$$

$$29. \operatorname{cosec} 67^\circ = \sec 23^\circ \rightarrow \sec^2 23^\circ - \tan^2 23^\circ = \mathbf{1}$$

$$30. \cos 54^\circ = \sin 36^\circ, \cos 36^\circ = \sin 54^\circ \rightarrow 1 + 1 = \mathbf{2}$$

$$31. \sin 61^\circ = \cos 29^\circ \rightarrow \sin^2 29^\circ + \cos^2 29^\circ = \mathbf{1}$$

Section D — Ratios from a right triangle

$$32. AC = \sqrt{3^2 + 4^2} = 5 \text{ cm. } \sin A = \mathbf{4/5}, \cos A = \mathbf{3/5}, \tan A = \mathbf{4/3}, \sin C = \mathbf{3/5}$$

$$33. QR = \sqrt{13^2 - 5^2} = 12 \text{ cm. } \tan P = \mathbf{12/5}, \tan R = \mathbf{5/12}$$

$$34. YZ = \sqrt{25^2 - 7^2} = 24 \text{ cm. } \sin Z = \mathbf{7/25}, \cos Z = \mathbf{24/25}$$

$$35. AB = \sqrt{15^2 + 20^2} = 25 \text{ cm. } \sin A = \mathbf{4/5}, \cos A = \mathbf{3/5}, \tan A = \mathbf{4/3}, \operatorname{cosec} A = \mathbf{5/4}, \sec A = \mathbf{5/3}, \cot A = \mathbf{3/4}$$

Section E — Challenge expressions

36. 1st bracket = $\sin 90^\circ = 1$, 2nd = $\cos 90^\circ = 0 \rightarrow 1^2 + 0^2 = \mathbf{1}$

38. $= (2\sqrt{3})/(4\sqrt{3}) = \sqrt{3}/2$ (= $\sin 60^\circ$)

40. $= (1 - 1)/(1 + 1) = 0/2 = \mathbf{0}$

37. $1 + \cot^2 30^\circ = \operatorname{cosec}^2 30^\circ = 4 \rightarrow 4 \times \frac{1}{4} = \mathbf{1}$

39. $= (2\sqrt{3})/2 = \sqrt{3}$ (= $\tan 30^\circ$)